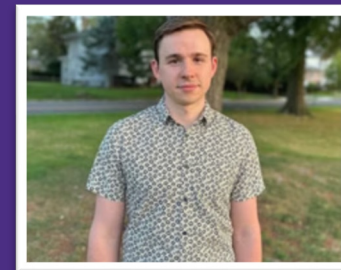
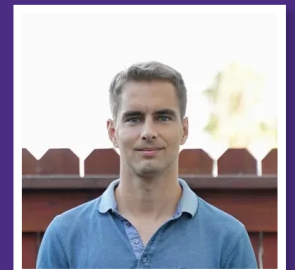


Spurions, Symmetry, and (RG) Stability



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Outline



- Motivation / Review
- Simple Supersoft Example
- General Classification Algorithm
- Output / Results (*Preliminary*)



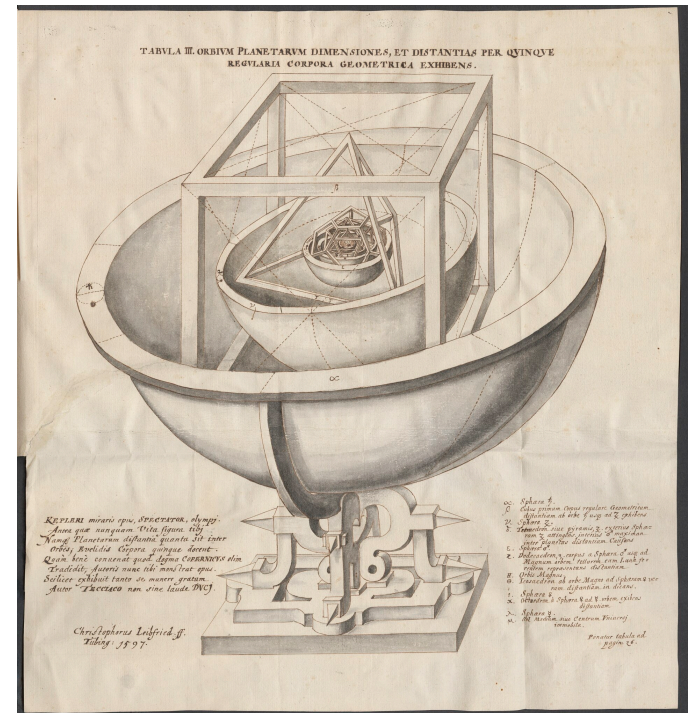
Fine-Tuning: An Odyssey

“We may ... look on the Universe as a symphony played on seven primitive constants as music played on the seven notes of a scale” – Sir Arthur Eddington

➤ Epicycles (*Ptolemy* ~150)

➤ *Mysterium Cosmographicum* (*Kepler* 1596)

➤ Titius-Bode Law (1766/1772) $a = 4 + 3 \times 2^n$



Naturalness & Renormalization Group (RG) Flow

- Dirac Naturalness (Dirac '38, Weisskopf '34, '39)

$$g_{\text{phys}} \sim \Lambda^d \mathcal{O}(1)$$

- Technical Naturalness (*Wilson '71, Weinberg '76, Susskind '79, Veltman '81*)

$$|\delta g_{\text{radiative}}| \lesssim |g_{\text{phys}}|$$

- 't Hooft Naturalness (*1980*)

$$\delta g_{\text{radiative}} \rightarrow 0 \Rightarrow \text{Symmetry Enhanced}$$

$$\beta_{m^2} \equiv \mu \frac{dm^2}{d \log \mu} = \underbrace{\alpha m^2}_{\text{technical / 't Hooft naturalness}} + \underbrace{\delta \Lambda_{\text{UV}}^2}_{\text{Dirac-natural expectation}}$$



Spurions

$$\mathcal{L} \supset \lambda \mathcal{O}, \quad \lambda \rightarrow \rho^*(g) \lambda, \quad \mathcal{O} \rightarrow \rho(g) \mathcal{O}, \quad g \in G$$

- Isospin $\Delta I = \frac{1}{2}$ Selection Rules (*D'Espagnat, Prentki, Salam '57*)
- Current Algebra / Chiral Lagrangian (*Gasser & Leutwyler '83*)
- Supersymmetry (SUSY) (*Girardello '82, Seiberg '93, Arkani-Hamed '98*)
- Effective Field Theory (*Wilson '75, Weinberg '79*)



A $\mathbb{Z}_2$ Complex Scalar Field (Partial)

$$\mathcal{L} = \partial_\mu \Phi^\dagger \partial^\mu \Phi - m_0^2 |\Phi|^2 - \frac{1}{4} \lambda_0 |\Phi|^4 \\ - \frac{1}{4} (m_1^2 \Phi^2 + \text{h. c.}) - \frac{1}{12} (\lambda_2 \Phi^2 + \text{h. c.}) |\Phi|^2 - \frac{1}{48} (\lambda_4 \Phi^4 + \text{h. c.})$$

Perturbative β Function at All Orders

$$\beta_{m_0^2} = a_0^{(0)} m_0^2 + a_2^{(1)*} m_1^2 + \text{c. c.} \\ \beta_{m_1^2} = b_0^{(0)} m_1^2 + b_4^{(1)} m_1^{2*} + b_2^{(2)} m_0^2$$

$$\therefore \\ m_1^2 = \lambda_2 = \lambda_4 = 0 \quad [U(1)] \\ m_0^2 = \lambda_2 = \lambda_4 = 0 \quad [\text{SSoft Scale } U(1)] \\ m_1^2 = \lambda_2 = 0 \quad [\mathbb{Z}_4] \\ m_0^2 = \lambda_2 = 0 \quad [\text{SSoft Scale } \mathbb{Z}_4]$$

[Fox, Nelson, Weiner '02,
Ferreira, et al '23, Haber, Ferreira '25]

1 Complex Scalar Field (Full)

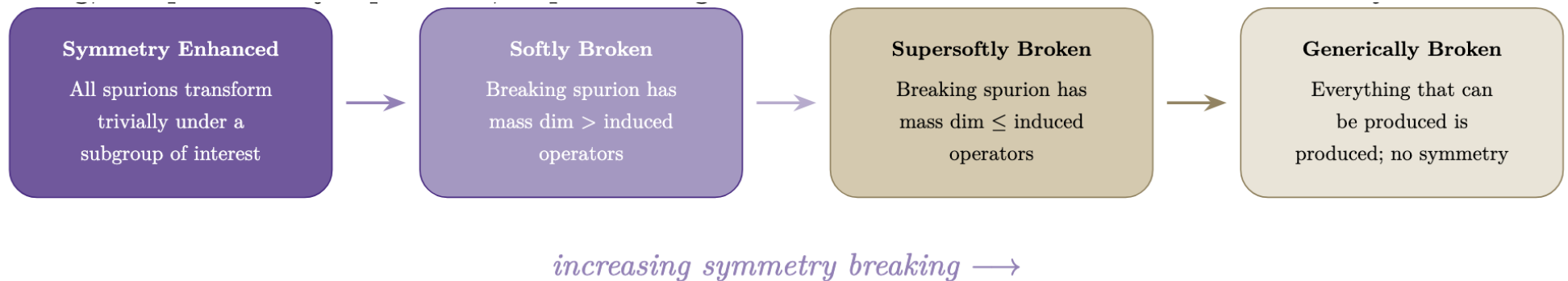
$$\begin{aligned}\mathcal{L} = & \partial_\mu \Phi^\dagger \partial^\mu \Phi - m_0^2 |\Phi|^2 - \frac{1}{4} \lambda_0 |\Phi|^4 \\ & - \frac{1}{4} (m_1^2 \Phi^2 + \text{h.c.}) - \frac{1}{4} (t_1 \Phi + \text{h.c.}) |\Phi|^2 - \frac{1}{12} (t_3 \Phi^3 + \text{h.c.}) \\ & - \frac{1}{12} (\lambda_2 \Phi^2 + \text{h.c.}) |\Phi|^2 - \frac{1}{48} (\lambda_4 \Phi^4 + \text{h.c.}),\end{aligned}$$

$$\begin{aligned}\beta_{m_0^2} &= a_0^{(0)} m_0^2 + a_2^{(1)*} m_1^2 + a_2^{(2)} |t_3|^2 + a_2^{(5)} t_1 t_3^* + a_4^{(6)*} t_1 t_3 + a_6^{(7)*} t_3^2 + \text{c.c.} \\ \beta_{m_1^2} &= b_0^{(0)} m_1^2 + b_4^{(1)} m_1^{2*} + b_2^{(2)} m_0^2 + b_0^{(5)} t_1^2 + b_4^{(6)*} t_3^2 + b_4^{(7)} t_1^{*2} + b_8^{(8)} t_3^{*2} \\ &\quad + b_2^{(9)*} t_1 t_3 + b_6^{(10)} t_1^* t_3^* + b_0^{(11)} t_1^* t_3^* \\ \beta_{t_1} &= c_0^{(0)} t_1 + c_2^{(1)} t_1^* + c_2^{(2)*} t_3 + c_4^{(1)} t_1 + d_4^{(2)} t_1^* + d_6^{(3)} t_3^* \\ \beta_{\lambda_0} &= e_0^{(0)}, \quad \beta_{\lambda_2} = f_2^{(0)}, \quad \beta_{\lambda_4}\end{aligned}$$

Identification of a Fixed Point

Let $\mathcal{S} = \{\text{Sp}_1, \text{Sp}_2, \dots, \text{Sp}_n\}$ be the complete set of spurions of an RG-stable theory with symmetry group G in $D = 4$ spacetime dimensions. Then $\mathcal{S}$ is closed under spurion multiplication. That is, for any two spurions $\text{Sp}_i, \text{Sp}_j \in \mathcal{S}$, their product decomposes as

$$\text{Sp}_i \otimes \text{Sp}_j = \bigoplus_k m_{ij}^k \text{Sp}_k, \quad m_{ij}^k \in \mathbb{Z}_{\geq 0},$$



... Enter The Hilbert Series ...

Mini-Intro on Hilbert Series

How many ways can we combine red balls  ? We write a generating function:

$$1 + \text{red ball} + \text{red ball red ball} + \text{red ball red ball red ball} + \cdots = \frac{1}{1 - \text{red ball}}$$

For two types of balls  and :

$$1 + \text{red ball} + \text{blue ball} + \text{red ball blue ball} + \text{red ball red ball} + \text{blue ball blue ball} + \cdots = (1 + \text{red ball} + \text{red ball red ball} + \cdots)(1 + \text{blue ball} + \text{blue ball blue ball} + \cdots) = \frac{1}{(1 - \text{red ball})(1 - \text{blue ball})}$$

We can *grade* by number of balls using a counter o :

$$1 + o(\text{red ball} + \text{blue ball}) + o^2(\text{red ball red ball} + \text{red ball blue ball} + \text{blue ball blue ball}) + \cdots = \frac{1}{(1 - o \text{ red ball})(1 - o \text{ blue ball})}$$

Assign a cost:  costs n_r dollars and  costs n_b dollars, tracked by the \$ on each ball:

$$1 + o(\text{\$}^{\text{red ball}} + \text{\$}^{\text{blue ball}}) + o^2(\text{\$}^{2n_r} + \text{\$}^{n_r} \text{\$}^{n_b} + \text{\$}^{2n_b}) + \cdots = \frac{1}{(1 - o \text{\$}^{n_r})(1 - o \text{\$}^{n_b})}$$

To find all combinations of total cost $n_c = xn_r + yn_b$, read off the coefficient of $(\$)^{n_c}$.

Truncated Covariant Hilbert Series

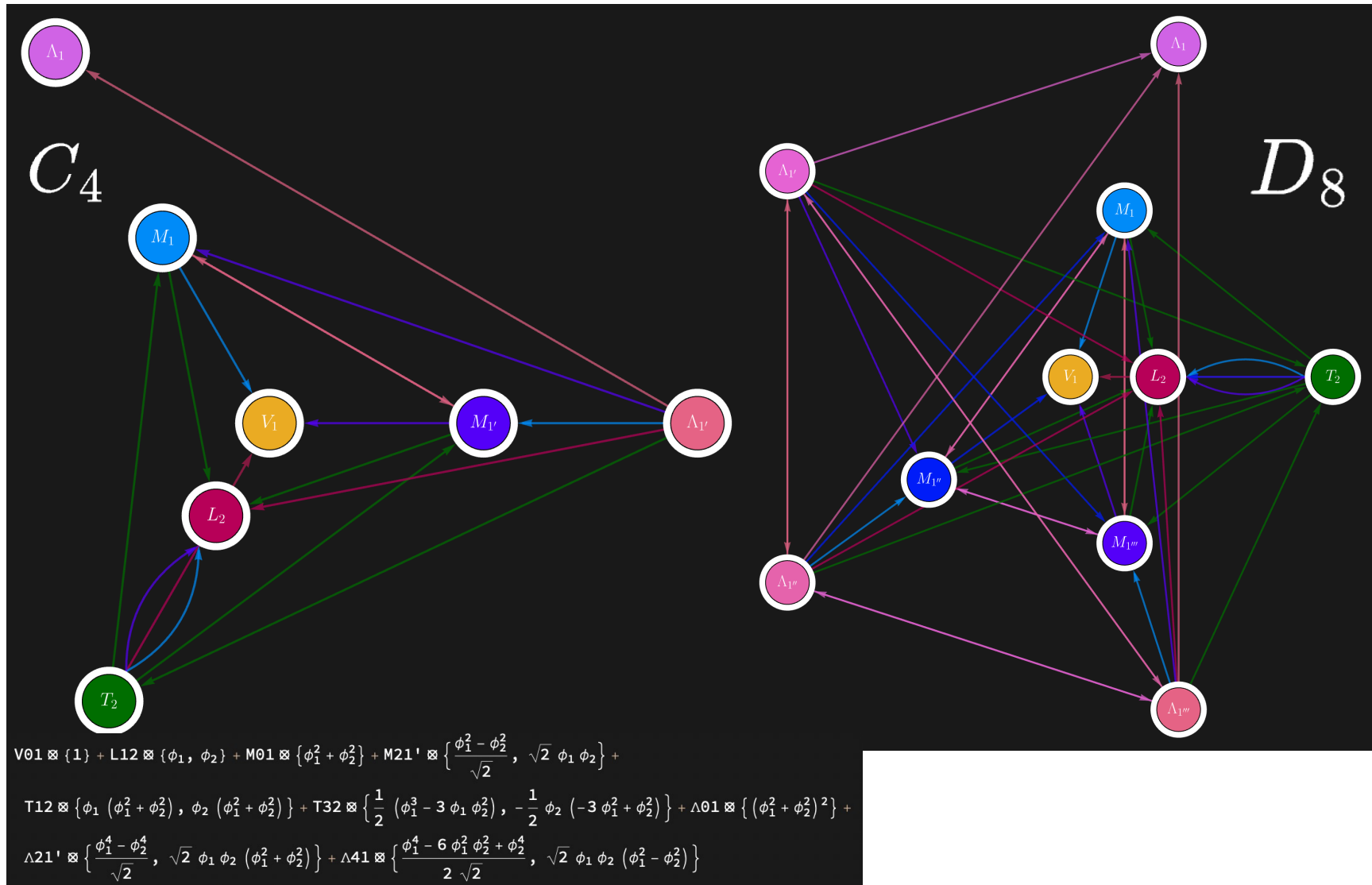
$$Z_G(\rho, \chi, n) = \left\{ \frac{1}{|G|} \sum_{\text{Cl}_G(g) \in \text{Cl}_G} |\text{Cl}_G(g)| \bar{\chi}(g) \text{PE}_n[\rho(g)] \right\}_{d^n}$$

$$\text{PE}_n[\rho(g)] = \prod_{\rho_i \in \rho} \prod_{j=1}^N \frac{1}{1 - o d^n \lambda_{\rho_i(g)}^{(j)} z_n(\rho_i)}$$



Detailed Group Data

Truncated Covariant Hilbert Series



Irreps and Eigenvalues

The Idempotent

$$[P_{\chi}^{\text{irrep}}(G, \rho)]_{ab} = \frac{\chi(1)}{|G|} \sum_{g \in G} \bar{\chi}(g) \rho_{ab}(g)$$

$$\rho_{ab}^{(r)}(g) = \lambda_a^{i_1, \dots, i_r} \left[\prod_{n=1}^r R_{i_n j_n}(g) \right] \lambda_b^{j_1, \dots, j_r}$$

+ Frobenius Schur Indicator

Finding Closed Sets: NextClosure

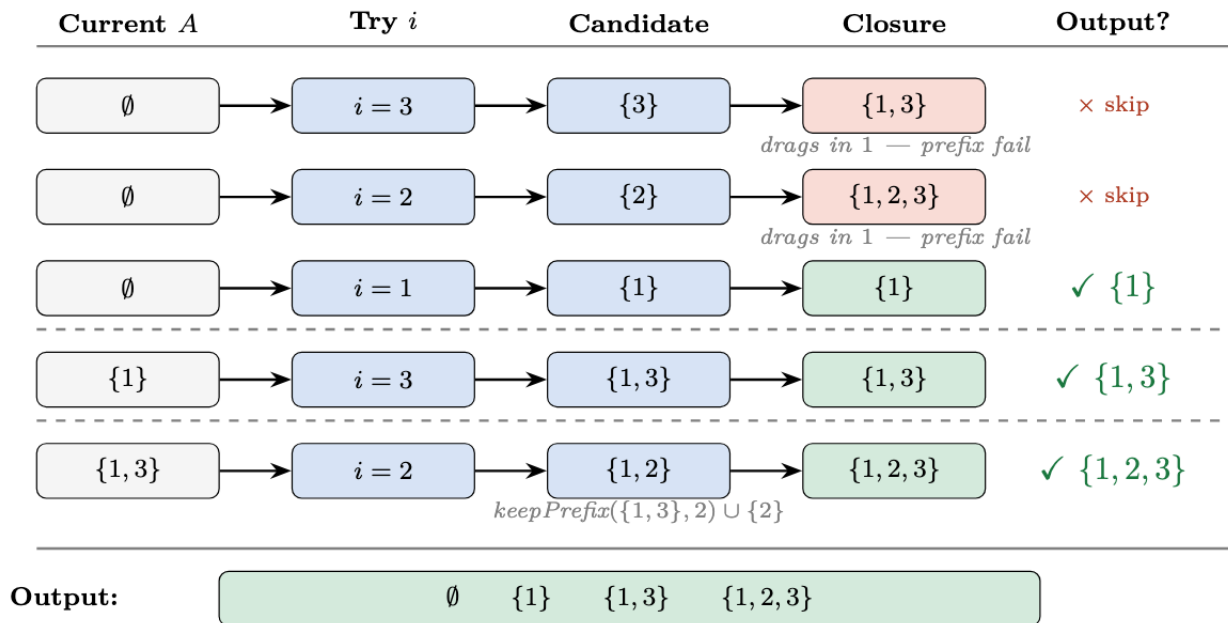
Multiplication Table

·	1	2	3
1	1	1,2,3	1,3
2	1,2,3	1,2,3	1,2,3
3	1,3	1,2,3	3

Entry ij shows what adding j to a set containing i forces in.

Closed sets:

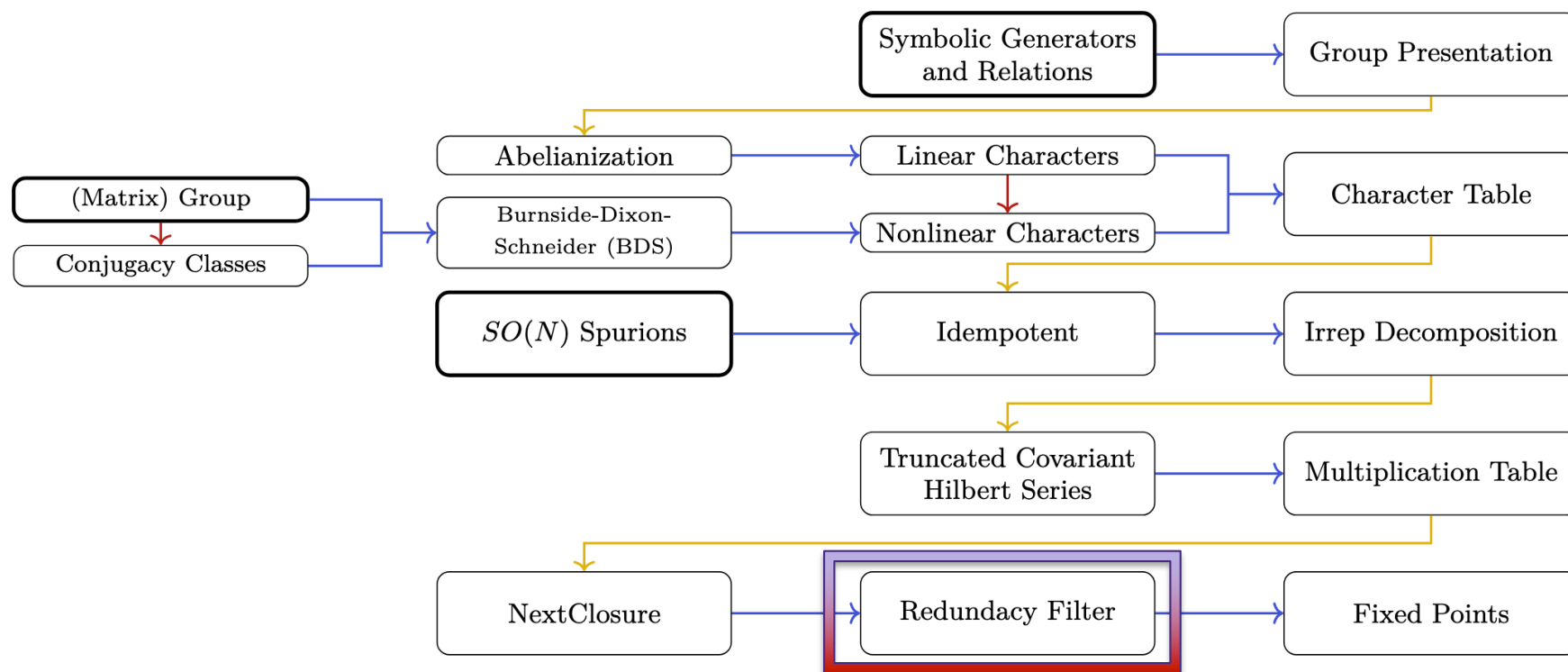
$\emptyset, \{1\}, \{1, 3\}, \{1, 2, 3\}$



- Green = prefix check passed, set accepted
- Red = closure drags in smaller element, skip

Key idea: at each step, try extending the current closed set by the next element in lexic order. If the closure respects the ordering, accept it. If it jumps ahead, skip it. This guarantees all closed sets are enumerated **exactly once**.

General Classification Algorithm



SO(2)/ O(2) / SO(3) Results

➤ 50 [24] O(2) [SO(2)] Fixed Points

➤ CP-Odd Basis Dependence

➤ 147 SO(3) Fixed Points

➤ 3-scalar Ssoft version

No.	m_0^2	m_1^2	t_1	t_3	λ_0	λ_2	λ_4	Type
1	0	0	0	0	0	0	0	Free massless
2	0	R	0	0	0	0	0	Free $\pm$ masses
3	0	0	0	0	0	0	0	Free $U(1)$
4	0	0	0	0	0	0	0	Scale $U(1)$
5	R	0	0	0	0	0	0	Free arbitrary mass
6	{0}	R	0	0	0	0	0	SSoft Scale $U(1)$
7	0	0	0	R	0	0	0	Special case of 16
8	0	0	0	0	0	R	0	Scale $\mathbb{Z}_4$
9	0	0	0	0	0	0	0	$U(1)$
10	R	0	$\text{Re}^{i\theta}$	0	0	0	0	Special case of 21
11	R	0	0	0	0	0	0	Soft $U(1)$
12	{0}	R	0	0	0	R	0	SSoft Scale $\mathbb{Z}_4 + \text{CP}$
13	R	R	0	0	0	0	0	
14	R	0	R	0	0	0	0	Special case of 10
15	0	0	0	0	0	0	0	Special case of 28
16	0	0	R	0	0	0	0	$\mathbb{Z}_3$ CP
17	0	0	0	0	R	R	0	Scale CP
18	0	0	0	0	0	R	0	Scale $\mathbb{Z}_4$
19	R	0	{0}	0	0	0	0	Special case of 31
20	R	0	0	0	0	0	0	Soft $\mathbb{Z}_3$
21	R	0	$\text{Re}^{i\theta}$	0	0	0	0	Soft ($\mathbb{Z}_3$ CP)
22	0	R	0	0	0	0	0	SSoft Scale $\mathbb{Z}_4$
23	R	0	0	0	0	R	0	(Soft Scale $\mathbb{Z}_4$) + CP
24	0	R	0	0	$i\text{R}$	R	0	SSoft Scale CP
25	R	R	R	0	0	0	0	CP no quartics
26	R	R	0	0	0	0	0	Special case of 31
27	R	0	R	0	0	0	0	Special case of 21
28	0	0	0	0	0	0	0	$\mathbb{Z}_3$
29		$\text{Re}^{i\theta}$	$\text{Re}^{i\theta}$	0	0	0	0	Special case of 38
30		$\text{Re}^{i\theta}$	$\text{Re}^{-i\theta}$	0	0	0	0	Special case of 39
31	R	0	{0}	0	0	0	0	SSoft
32	R	0	0	0	0	0	0	Soft $\mathbb{Z}_3$
33	R	0	0	R	R	0	0	Soft CP
34	R	0	0	0	0	0	0	Soft $\mathbb{Z}_4$ (1)
35	R	R	R	0	0	0	0	
36	0	0	0	0	R	0	0	Scale
37	R	0	0	0	0	0	0	
38		$\text{Re}^{i\theta}$	$\text{Re}^{i\theta}$	0	0	0	0	Special case of 43
39		$\text{Re}^{i\theta}$	$\text{Re}^{-i\theta}$	0	0	0	0	Special case of 44
40	0	0	0	R	R	0	0	Soft ($\mathbb{Z}_2$ CP)
41	R	R	R	0	R	0	0	
42	R	0	0	0	0	0	0	
43		R	R	0	R	0	0	Special case of 44
44		$\text{Re}^{i\theta}$	$\text{Re}^{-i\theta}$	0	R	0	0	
45	R	R	R	R	R	0	0	CP
46		$\text{Re}^{i\theta}$	$\text{Re}^{i\theta}$	R	R	0	0	
47	R	0	0	0	0	0	0	$\mathbb{Z}_2$
48	R	0	0	0	0	0	0	Soft $\mathbb{Z}_4$ (2)
49					R	R	0	
50	R							Full theory

A Tantalizing Hint

$$\begin{aligned}
\mathcal{L} = & -|\partial H|^2 - \mu_0^2 |H|^2 - \frac{g_0}{4} |H|^4 \\
& - \left[\frac{M_1^2}{4} H^2 + \text{c.c.} \right] - \left[\frac{g_2}{12} H^2 |H|^2 + \text{c.c.} \right] - \left[\frac{g_4}{48} H^4 + \text{c.c.} \right] \rightarrow \text{forbidden by } U(1)_Y \text{ if hypercharged} \\
& - |\partial \Phi|^2 - m_0^2 |\Phi|^2 - \frac{\lambda_0}{4} |\Phi|^4 \\
& - \left[\frac{m_1^2}{4} \Phi^2 + \text{c.c.} \right] \\
& - \left[\frac{t_1}{4} \Phi |\Phi|^2 + \text{c.c.} \right] - \left[\frac{t_3}{12} \Phi^3 + \text{c.c.} \right] \rightarrow \text{forbidden by } \mathbb{Z}_2 \\
& - \left[\frac{\lambda_2}{12} \Phi^2 |\Phi|^2 + \text{c.c.} \right] - \left[\frac{\lambda_4}{48} \Phi^4 + \text{c.c.} \right] \\
& - [A_1 \Phi |H|^2 + \text{c.c.}] - g_0^c |\Phi|^2 |H|^2 - \left[\frac{g_2^c}{4} \Phi^2 |H|^2 + \text{c.c.} \right] \\
& + U(1)_Y\text{-violating portals: } \Phi H^2, \Phi^2 H^2, |\Phi|^2 H^2
\end{aligned}$$

Assume H carries hypercharge and Φ obeys $\mathbb{Z}_2$ symmetry $\Rightarrow t_{1,3} = A_1 = 0$.

SuperSoft RG fixed point:

$$\begin{cases} \mu_0^2 = m_0^2 = 0 ; & m_1^2, \mu_0^2 \text{ arbitrary} \\ \lambda_2 = g_2^c = 0 ; & g_0, g_0^c, \lambda_4, \lambda_0 \text{ arbitrary} \end{cases}$$

No hierarchy problem for $m_1^2 \gg \mu_0^2, m_0^2$, but a large “classical” mass.

Caveats for the Critic

- UV Sensitivity (Non-Wilsonian)
- No Dimensionless Fixed Points (e.g. Banks-Zaks)
- Group-Theory Focused (Upper Triangular, Basis Choices)

Conclusions and Future Directions

Invariance and covariance together yield RG stability,
systematically characterized by a spurion analysis.

Additional Directions

- Fermions
- UV Completions
- Non-Renormalizable Theories
- Finite Loop Order
- SUSY
- Hierarchy Problems
- Non-Linear Realizations
- Non-Perturbative Statements

Further Analysis And Public Code (FixedPointsPlethysm) Forthcoming...

Thank You!

Renormalizable SO(N) Lagrangian

$$\begin{aligned}\mathcal{L}_{(4)} &= V_1 + L_N^i \Phi_i + (M^2)^{ij}_{N(N+1)/2} \Phi_i \Phi_j + T^{ijk}_{N(N+1)(N+2)/6} \Phi_i \Phi_j \Phi_k + \Lambda^{ijkl}_{N(N+1)(N+2)(N+3)/24} \Phi_i \Phi_j \Phi_k \Phi_l \\ &= V_1 + L_N^i \Phi_i + M_1^2 |\Phi|^2 + (M^2)^{ij}_{[N(N+1)/2]-1} \Phi_i \Phi_j + T_N^i \Phi_i |\Phi|^2 + T^{ijk}_{[N(N+1)(N+2)/6]-N} \Phi_i \Phi_j \Phi_k \\ &\quad + \Lambda_1 |\Phi|^4 + \Lambda^{ij}_{[N(N+1)/2]-1} \Phi_i \Phi_j |\Phi|^2 + \Lambda^{ijkl}_{[N(N+1)(N+2)(N+3)/24]-[N(N+1)/2]} \Phi_i \Phi_j \Phi_k \Phi_l,\end{aligned}$$



Alternate Spacetime Dimensions

$$\begin{aligned}\mathcal{L}_{(3)} = & V_1 + L_{\mathbf{N}}^i \Phi_i + (M^2)_{\mathbf{N}(\mathbf{N}+1)/2}^{ij} \Phi_i \Phi_j + T_{\mathbf{N}(\mathbf{N}+1)(\mathbf{N}+2)/6}^{ijk} \Phi_i \Phi_j \Phi_k + \Lambda_{\mathbf{N}(\mathbf{N}+1)(\mathbf{N}+2)(\mathbf{N}+3)/24}^{ijkl} \Phi_i \Phi_j \Phi_k \Phi_l \\ & + Q_{\mathbf{N}(\mathbf{N}+1)(\mathbf{N}+2)(\mathbf{N}+3)(\mathbf{N}+4)/120}^{ijklm} \Phi_i \Phi_j \Phi_k \Phi_l \Phi_m + S_{\mathbf{N}(\mathbf{N}+1)(\mathbf{N}+2)(\mathbf{N}+3)(\mathbf{N}+4)(\mathbf{N}+5)/720}^{ijklmn} \Phi_i \Phi_j \Phi_k \Phi_l \Phi_m \Phi_n\end{aligned}\quad (\text{A4})$$

$$\begin{aligned}= & V_1 + L_{\mathbf{N}}^i \Phi_i + M_1^2 |\Phi|^2 + (M^2)_{[\mathbf{N}(\mathbf{N}+1)/2]-1}^{ij} \Phi_i \Phi_j + T_{\mathbf{N}(\mathbf{N}+1)(\mathbf{N}+2)/6-\mathbf{N}}^i \Phi_i |\Phi|^2 + T_{[\mathbf{N}(\mathbf{N}+1)(\mathbf{N}+2)/6]-\mathbf{N}}^{ijk} \Phi_i \Phi_j \Phi_k \\ & + \Lambda_1 |\Phi|^4 + \Lambda_{[\mathbf{N}(\mathbf{N}+1)/2]-1}^{ij} \Phi_i \Phi_j |\Phi|^2 + \Lambda_{[\mathbf{N}(\mathbf{N}+1)(\mathbf{N}+2)(\mathbf{N}+3)/24]-[\mathbf{N}(\mathbf{N}+1)/2]}^{ijkl} \Phi_i \Phi_j \Phi_k \Phi_l \\ & + Q_{\mathbf{N}}^i \Phi_i |\Phi|^4 + Q_{[\mathbf{N}(\mathbf{N}+1)(\mathbf{N}+2)/6]-\mathbf{N}}^{ijk} \Phi_i \Phi_j \Phi_k |\Phi|^2 + Q_{[\mathbf{N}(\mathbf{N}+1)(\mathbf{N}+2)(\mathbf{N}+3)(\mathbf{N}+4)/120]-[\mathbf{N}(\mathbf{N}+1)(\mathbf{N}+2)/6]}^{ijklm} \Phi_i \Phi_j \Phi_k \Phi_l \Phi_m \\ & + S_1 |\Phi|^6 + S_{[\mathbf{N}(\mathbf{N}+1)/2]-1}^{ij} \Phi_i \Phi_j |\Phi|^4 + S_{[\mathbf{N}(\mathbf{N}+1)(\mathbf{N}+2)(\mathbf{N}+3)/24]-[\mathbf{N}(\mathbf{N}+1)/2]}^{ijkl} \Phi_i \Phi_j \Phi_k \Phi_l |\Phi|^2 \\ & + S_{[\mathbf{N}(\mathbf{N}+1)(\mathbf{N}+2)(\mathbf{N}+3)(\mathbf{N}+4)(\mathbf{N}+5)/720]-[\mathbf{N}(\mathbf{N}+1)(\mathbf{N}+2)(\mathbf{N}+3)/24]}^{ijklmn} \Phi_i \Phi_j \Phi_k \Phi_l \Phi_m \Phi_n\end{aligned}\quad (\text{A5})$$

$$\begin{aligned}\mathcal{L}_{(5)/(6)} = & V_1 + L_{\mathbf{N}}^i \Phi_i + (M^2)_{\mathbf{N}(\mathbf{N}+1)/2}^{ij} \Phi_i \Phi_j + T_{\mathbf{N}(\mathbf{N}+1)(\mathbf{N}+2)/6}^{ijk} \Phi_i \Phi_j \Phi_k \\ = & V_1 + L_{\mathbf{N}}^i \Phi_i + M_1^2 |\Phi|^2 + (M^2)_{[\mathbf{N}(\mathbf{N}+1)/2]-1}^{ij} \Phi_i \Phi_j + T_{\mathbf{N}(\mathbf{N}+1)(\mathbf{N}+2)/6-\mathbf{N}}^i \Phi_i |\Phi|^2 + T_{[\mathbf{N}(\mathbf{N}+1)(\mathbf{N}+2)/6]-\mathbf{N}}^{ijk} \Phi_i \Phi_j \Phi_k,\end{aligned}\quad (\text{A6})$$



N-Higgs Doublet

$$\mathcal{L} = -\partial^\mu \Phi^\dagger \partial_\mu \Phi - M_0^2 |\Phi|^2 + \Lambda_0 |\Phi|^4 + M_i^2 (\Phi^\dagger \sigma_i \Phi) + \Lambda_i (\Phi^\dagger \sigma_i \Phi) |\Phi|^2 + \Lambda_{ij} (\Phi^\dagger \sigma_i \Phi) (\Phi^\dagger \sigma_j \Phi)$$

$$\beta_{M_0^2} = a_0^{(0)} M_0^2 + a_i^{(1)} M_i^2, \quad \beta_{M_i^2} = b_0^{(0)} M_i + b_{ij}^{(1)} M_j^2 + b_i^{(2)} M_0^2$$

$$\beta_{\Lambda_0} = c_0, \quad \beta_{\Lambda_i} = d_i^{(0)}, \quad \beta_{\Lambda_{ij}} = e_{ij}^{(0)}$$

